Energy 256 (2022) 124612

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

SOC estimation of Li-ion battery using convolutional neural network with U-Net architecture

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ARTICLE INFO

Article history:
Received 24 February 2022
Received in revised form
20 May 2022
Accepted 20 June 2022
Available online 23 June 2022

Keywords:
SOC estimation
Neural network
CNN
U-Net
Total variation
Bayesian optimization

ABSTRACT

State-of-charge (SOC) is critical to the safe operation and energy management of electric vehicles. Data-driven SOC estimation algorithms all require a period of data to ensure convergence of the estimation results and cannot accurately estimate the SOC values near the starting point. We propose a SOC estimation method based on the U-Net architecture that can handle variable-length input data and output equal-length SOC estimation results, including accurate SOC of the starting point. Symmetric padding convolutional layer was proposed to address the boundary effect of Convolutional Neural Networks (CNN) and improve the accuracy of SOC estimation at the edges. We also propose a total variation loss function, which improves the stability of the estimation only by optimizing the loss function without increasing the model complexity, and significantly reduces the maximum error. The model was trained using dynamic drive cycle data at five constant temperatures, and the model has high accuracy at both constant and variable temperature conditions. The proposed method can estimate the SOC at constant temperatures with mean absolute error (MAE) within 1.1% and root-mean-square error (RMSE) within 1.4%. This method also can estimate SOC at varying temperatures with MAE within 1.5% and RMSE within 1.8% under different driving conditions.

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1. Introduction

Lithium batteries are increasingly used in vehicles due to their high energy density and low environmental pollution characteristics. In order to monitor battery status, predict remaining mileage and optimize energy dispatch, electric vehicles use battery management systems (BMS) for battery monitoring and safety protection. The most important and core parameter for BMS is the state of charge (SOC) [1]. Accurate SOC estimation can improve battery safety, prevent abnormal charging and discharging, improve driving range, and relieve mileage anxiety [2]. However, SOC is a quantity that cannot be measured directly and has strong nonlinearity [3]. We can only estimate SOC indirectly from the current and voltage of the battery, so an accurate SOC estimation is a core requirement of the BMS [4,5].

Over the years, scholars have conducted comprehensive research on battery SOC estimation, and many papers have been unanimously recognized. In general, these methods can be divided into four categories: ampere-time integration methods [6], open-circuit voltage methods [7], model-based methods [8] and data-driven methods [9]. Ampere-time integration method is the most commonly used method in BMS because of its small calculation volume and high accuracy, which can be conveniently implemented on BMS chips with low calculation power. However, the accuracy of the ampere-time integration method is affected by the initial SOC, and the temperature drift and noise of the current sensor can lead to cumulative errors in the integration process [10]. The open-circuit voltage method can obtain the remaining capacity of the battery, but this method requires the battery to rest for a long time until the terminal voltage is fully depolarized before SOC estimation [11,12]. Model-based SOC estimation methods have been developed and are widely used and generally requires an equivalent circuit model or electrochemical model of the battery. By introducing factors such as temperature and aging, the accuracy of the equivalent circuit model can be improved, which in turn improves the accuracy of SOC estimation [13]. However, these methods need to construct an accurate model of the battery and require the measurement of numerous parameters, some of which are not directly available [14,15]. The data-driven approach treats the battery system as a black box and uses only external observable

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https://doi.org/10.1016/j.energy.2022.124612
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Nomenclature

symbols

α	weighting factor
$\hat{y}_k$	estimated SOC at data point k
λ	regularization weight
μ	mean of measurement data
σ	standard deviation of measurement data
$loss$	total loss function
$loss_{mae}$	mean absolute error loss function
$loss_{tv}$	total variation loss function
x	measurement data
x_k	measurement at data point k
x_{std}	standardized measurement data

subscripts

k	k-th data point
mae	mean absolute error
std	standard deviation
tv	total variation

abbreviations

BMS	Battery Management System
BP	Back Propagation
BTMS	Battery Thermal Management System
CNN	Convolutional Neural Network
ECM	Equivalent Circuit Model
EKF	Extended Kalman Filter
EV	Electric Vehicle
FC	Fully Connected
GRU	Gated Recurrent Unit
HEV	Hybrid Electric Vehicle
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MSE	Mean Square Error
NN	Neural Network
OCV	Open Circuit Voltage
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
SOC	State of Charge

data for SOC estimation without considering the internal mechanism, *e.g.,* random forest trees, support vector machines [16,17]. The advantage of this approach is that it does not need to measure or model the battery. The disadvantage of this approach is that the estimation results are heavily dependent on the training data, which makes it difficult to extrapolate. Accurate battery state estimation, including thermal estimation and SOC estimation, can improve the safety and economy of the battery pack [18,19].

In recent years, with the development of artificial neural networks, deep learning methods have been applied in the battery field [20–23]. Compared with traditional machine learning methods, deep learning can extract more hidden features and thus improve the accuracy of the model. Dong et al. [24] used back propagation (BP) neural network to predict SOC of a battery based on the current, voltage and temperature, but they only verified the accuracy under constant current charging. Xu et al. [25] used a stochastic fuzzy neural network (SFNN) instead of the traditional neural network that has filtering effect on noisy input to model the battery nonlinear dynamic. The disadvantage of neural networks is their unstable output, which is highly volatile compared to model-based estimation methods. To solve this problem some scholars use neural networks in combination with other traditional methods. Zhang et al. [26] introduced a particle filtering process after the BP neural network, which greatly improves the stability of the output results. Charkhgard et al. [27] used neural networks jointly with Kalman filtering to improve the estimation accuracy. Chen et al. [28] proposed a synthetic SOC estimation method for lithium-ion batteries based on long short-term memory network (LSTM) modeling and adaptive H-infinity filter. In their conclusion, the maximum SOC error of the LSTM network output after H-infinity filtering was reduced by 90%. Purohit et al. designed soft sensors to estimate the state of charge, state of energy, and power loss, and achieved simultaneous estimation of multiple states through neural networks [29].

Although neural networks (NN), especially recurrent neural networks (RNN), have significant advantages in SOC estimation, they also have a serious drawback that they require a long period of historical data sequence to reach a stable output [30,31]. Accurate SOC estimation is not possible for the beginning of a period of data, and there is a convergence-like process [32]. In the process of data analysis we often want to estimate the SOC for all points in the entire time period. Using RNN requires discarding the SOC estimation results at the beginning stage because the output is not yet stable [33]. If there is no accurate initial SOC, the RNN requires a period of data for the estimated results to converge. The length of historical data to make the output reach the accuracy requirement varies depending on the network topology, from tens of seconds to hundreds of seconds, which affects our analysis of the state of the starting phase of the battery. Meanwhile, these networks have fully connected layers inside, which leads to the fact that the input data length can only be fixed values, and the data used in the estimation process must be the same length as the training process. If the estimated data is longer than the training data, the data needs to be cropped; if the estimated data is shorter than the training data, it will result in the inability to use the trained model for SOC estimation.

To address the above problems, we propose a pure convolutional neural network-based SOC estimation method. The output SOC length of our proposed method is the same as the input data length, and the SOC estimation can be performed for each point without the convergence process. The output results maintain high accuracy at all points, including the starting point. Also, our method uses only convolutional neural network (CNN) architecture, which enables variable length of input and output data, including the training and estimation processes. To improve the accuracy and reduce the volatility of the estimated SOC, we add a total variation loss function penalty to the training process of the network, which improves the stability of the network output without increasing complexity.

The rest of this paper is organized as follows: Section II presents our CNN-based SOC estimation network using the U-Net architecture and compares its advantages. Section III describes the dataset we used and illustrates the improvements we made to enhance the stability of our proposed model. Our improvements include symmetric padding convolutional layers, total variation loss function penalty and Bayesian hyperparameter optimization. Section IV shows the SOC estimation results of our proposed model, including various drive cycle scenarios at constant and variable temperature. Finally, Section V summarizes the conclusions.

2. Proposed SOC estimation network based on convolutional neural network

2.1. Motivation behind CNN-based SOC estimation

Neural network SOC estimation methods proposed in previous literature use a segment of battery measurement data, typically including current, voltage and temperature to estimate the SOC. In order to obtain accurate estimation results, the neural network has certain requirements on the length of the input data. If the data length is insufficient, it will lead to a decrease in the stability of the estimation results. This SOC estimation method can be described with equation:

$$\hat{y}_{k+m} = f(x_{k+1}, x_{k+2}, \ldots, x_{k+m}) \tag{1}$$

where $x_k = [I_k, V_k, T_k]$ represents the battery measurement, including the current, voltage and temperature. $\hat{y}_{k+m}$ represents the SOC estimation result at timestamp $k + m$. The length of the input data required by a neural network is m, which is related to the neural network design. If m is small, the output stability is lower, if m is large, the length of the estimated output is short. Thus, we cannot accurately estimate the SOC near the starting point. We call this method "segment to point".

We proposed "segment to segment" SOC estimation method. This method uses the entire battery data as input to estimate the SOC of each point in the entire time period, which can be described as:

$$[\hat{y}_{k+1}, \hat{y}_{k+2}, \ldots, \hat{y}_{k+m}] = f(x_{k+1}, x_{k+2}, \ldots, x_{k+m}) \tag{2}$$

With this idea we build a neural network using convolutional layers and estimate the SOC for each input point at once, including the starting point. Fig. 1 shows the difference between the two methods. The original data length is the same for both methods, but due to the structural differences, the first method results in a portion of the SOC data that cannot be estimated, i.e., the dashed part in Fig. 1 (a).

2.2. SOC estimation network with U-Net architecture

We designed a CNN with U-Net architecture for SOC estimation. The U-Net architecture stems from the so-called "fully convolutional network" [34]. This network architecture was born in the field of computer vision and targets the segmentation task of biomedical images. While traditional image recognition tasks can only classify an image as a whole, U-Net can perform classification prediction for each pixel in the image to achieve accurate semantic segmentation. The convolutional layer is the core building block of a CNN, and it is where the majority of computation occurs. The one-dimensional convolution can be described by (3). I represents the input data, K represents the kernel, and F represents the feature map, which is the output of the convolution operation. m represents the indexes of the feature map and n represents the indexes of the kernel, respectively.

$$F_m = (I*K)_m = \sum_n K_n I_{m-n} \tag{3}$$

The U-Net architecture is designed for the semantic segmentation task, where the input is a two-dimensional picture and the output is in essence a classification task which is performed for each pixel in the image. We referenced this idea and designed a U-Net network for battery SOC estimation, where the input data becomes temporal one-dimensional battery measurement data and the output is changed to a regression task, i.e., SOC estimation. We also make a series of improvements on our proposed method for battery data to improve the continuity and stability of the estimation results. The structure of the SOC estimation network is shown in Fig. 2. The left side is the input data, which contains the current, voltage and temperature measurement data for a period of time. One type of data is called a channel. In the paper, we use three types of data: current, voltage and temperature, *i.e.*, the input data is one-dimensional and has three channels. The right side is the output data, which is the SOC estimation result for a period of time. The output data is one-dimensional and has only one channel. The red circles in the figure are downsampling block, which contain convolution and pooling layers. During the downsampling process, the temporal information decreases and the feature information increases. After each downsampling, the perceptual field of the network expands, and more features can be extracted. This multi-layer convolutional struct can extract information at multiple time scales from local to global. The blue circles are the upsampling blocks, which contain the convolution layer, pooling layer and Upsampling layer. During the upsampling process, the feature information is gradually restored to temporal information until it is the same size as the input data. The blocks in each row of the network have the same output size, with blocks closer to the

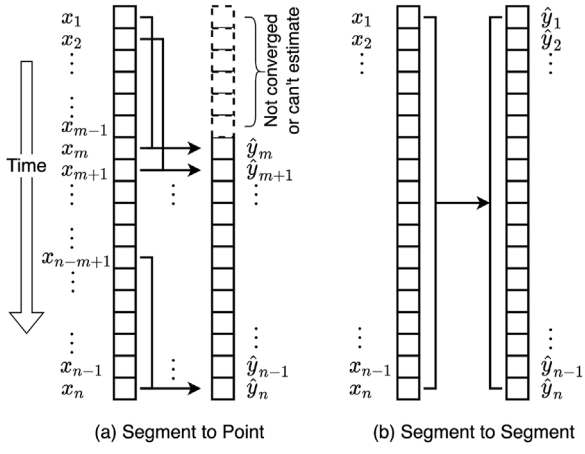


Fig. 1. Two different SOC estimation methods. (a) Segment to point, (b) segment to segment.

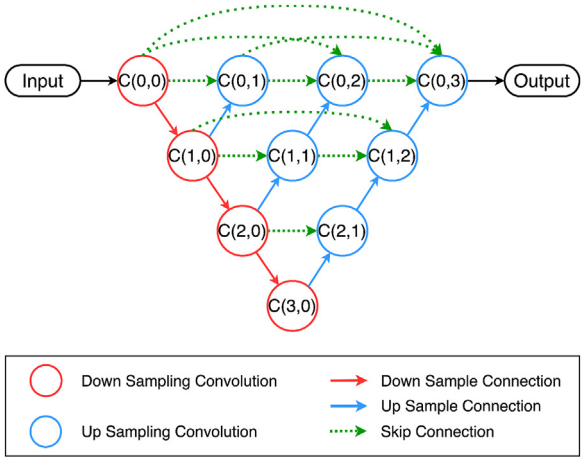


Fig. 2. SOC estimation network with U-Net architecture.

Table 1
Details of the implementation of U-Net.

Nodes	Structure
C(0,0) C(0,1) C(0,2) C(0,3)	[Conv(5,128) BatchNormalization(128)] × 3
C(1,0) C(1,1) C(1,2)	[Conv(5,256) BatchNormalization(256)] × 3
C(2,0) C(2,1)	[Conv(5,384) BatchNormalization(384)] × 3
C(3,0)	[Conv(5,512) BatchNormalization(512)] × 3

bottom having smaller sizes but more depth. The blocks in each row of the network have the same output size, with blocks closer to the bottom having smaller sizes but more depth. The green dashed lines represent skip connections, which can be used to concatenate blocks of the same size for better feature extraction.

The details of the U-Net we constructed are described in detail in the Table 1. The first column of the table indicates the names of the nodes that have the right-hand side structure, and since all nodes in each row have the same structure, they are placed together. Conv(5,128) represents a convolutional layer with kernel size of 5 and layer number of 128. BatchNormalization(128) represents a batch normalization layer with input and output sizes of 128. Multiplying by 3 means repeating this convolutional normalization process 3 times.

2.3. Main advantages of our proposed method

The most basic and widely used neural network is the fully connected (FC) layer, where each node of a layer is connected to all the nodes of the next layer [35]. Many problems can be solved using only the FC layer, such as the classic handwritten digit recognition problem also known as MNIST. Many SOC estimation network architectures include FC layers, but FC layer has the disadvantage that the input and output dimensions are fixed, which are determined at the construction stage of the network [33,36]. After the training process is over, the input and output dimensions of the entire network are also fixed, which affects the generalizability. CNN has the advantage that the input and output dimensions are variable. Each convolutional layer has two parameters: weights and biases, and the number of parameters is only related to the kernel number. This feature allows CNN to handle data of variable length. However, in a neural network architecture, as long as there is one FC layer, the input and output dimensions of the network can only be fixed values. Shelhamer et al. [37] first proposed a neural network consisting of only convolutional layers in 2014, known as a fully convolutional network (FCN).

In the paper we proposed a SOC estimation method draws on the core idea of FCN, using only convolutional structure to build neural network, enabling this method to handle battery measurement data of variable length. Our proposed method contains convolutional layers, pooling layers, batch normalization layers and upsampling layers. All four layers can handle arbitrary input and output dimensions. The symmetric structure makes the output to have the same length as the input. This feature allows our neural network to estimate the SOC of all data points simultaneously, without the need for a recursive process of point-by-point nor a convergence process. Another benefit of the CNN network is the fast computation speed, compared to the FC layer, where each node in an FC layer is connected to all nodes in the next layer, which contains numerous trainable parameters. In contrast, only kernel in CNN has trainable parameters. This difference in the number of parameters is more obvious when the dimensionality of the input data is large. Simultaneous computation of all data points also increases computational speed and reduces computational costs.

3. Experiment setup

3.1. Public dataset - Panasonic 18650 PF

We used a public LiB battery test dataset to validate proposed method [38]. The dataset used a brand new 2.9 Ah Panasonic 18650 PF cell. This battery has a nominal voltage of 3.6 V. The battery cell uses lithium nickel manganese cobalt oxide in its positive electrode (cathode), carbon material in its negative electrode (anode). C-rates are the rate of discharge (or charge) as compared to the capacity of the battery. The C-rate is often used in experiments to represent the current magnitude. The maximum current in this dataset is around 6C. The data collected include experiments at 5 different temperature from −20 °C to 25 °C. For each test the battery cell was fully charged and then was executed a drive cycle profile. There are four standard drive cycles included in the dataset, which are: Urban Dynamometer Driving Schedule (UDDS), Highway Fuel Economy Driving Schedule (HWFET), Los Angeles 92 (LA92) and Supplemental Federal Test Procedure Driving Schedule (US06). There are also five manually created drive cycles included in the dataset, which are random combinations of four standard drive cycles. The dataset contains current, voltage, temperature and ampere hour of the battery, all sampled at 10 Hz. Before each drive cycle at each temperature, the battery was fully charged constant current constant voltage at room temperature. It is reasonable to assume that the battery is 100% SOC before each experiment. The true SOC is calculated using the nominal capacity and the ampere hour value. We use the nominal capacity at 25 °C to calculate the SOC at different temperatures, which ensures that the battery has exactly the same SOC at different temperatures when it is not being charged or discharged. Fig. 3 shows one drive profile data at 25 °C ambient temperature. The SOC of the battery has a strong nonlinearity in this operating condition.

To improve the model training efficiency and increase the model accuracy, we standardized the input data. We set the mean of the input data to 0 and the standard deviation of the input data to 1. The standardization process is applied to all 5 temperature data in the dataset, since the output of the NN is the SOC estimation result, which is already in the [0,1] interval, no standardization is required. The equation used for standardization is (4), where x is the original

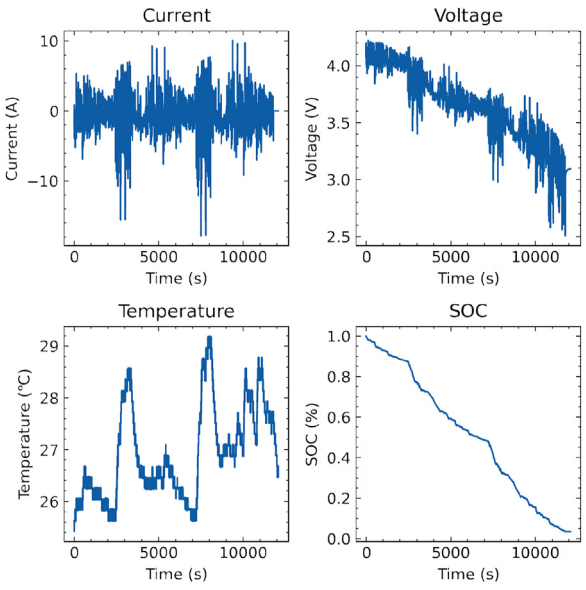


Fig. 3. One drive cycle data at an ambient temperature of 25 °C.

data, x_{std} is the standardized data, μ is the mean value, and σ is the standard deviation.

$$x_{std} = \frac{x - \mu}{\sigma} \tag{4}$$

3.2. Symmetric padding convolutional layer

CNN is a class of artificial neural network. They are also known as shift invariant or space invariant artificial neural networks, based on the shared-weight architecture of the convolution kernels or filters that slide along input features and provide translation equivariant responses known as feature maps. Convolutional computation leads to a reduction in the size of the output result. Taking the one-dimensional convolution calculation as an example, the length of the input is w and the length of the kernel is k, the length of the output will be $w - k + 1$. Thus, the data shrinks every time a convolution operation is preformed. This puts a limit on the number of convolutional layers and prevents us from building deeper networks. To solve this problem we can use padding, which is simply a process of adding additional zeros to the input data. The output will be the same size as the input if we add $(k - 1)/2$ zeros to the borders. This is called "zero padding" [39]. By zero padding we can solve the problem of size shrinking during convolution, but this method will affect the accuracy of the estimation results at the borders, because we are padding misleading information. The phenomenon that accuracy degrades at the edges of the data is called boundary effect.

In order to improve the accuracy of SOC estimation at the borders, we proposed a symmetric padding convolutional layer. The convolutional layer is similar to the zero padding one, but instead of adding zeros to the left and right, we mirror the data at the borders. This is called "symmetric padding". Fig. 4 shows three different padding methods. In the case of no padding, the output size is smaller than the input size, while in the case of zero padding and symmetric padding, the output is the same size as the input. The difference between zero padding and symmetric padding is that

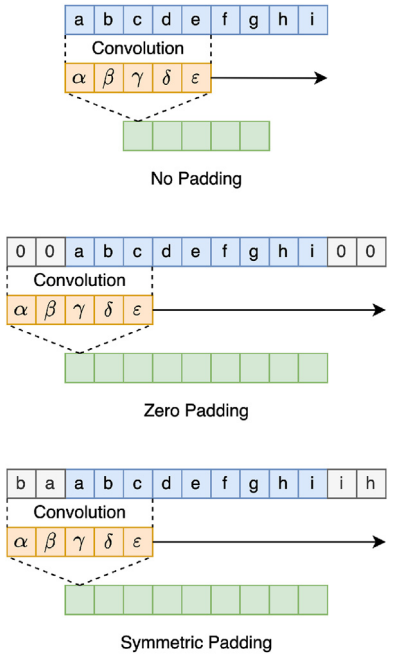


Fig. 4. Three different padding methods.

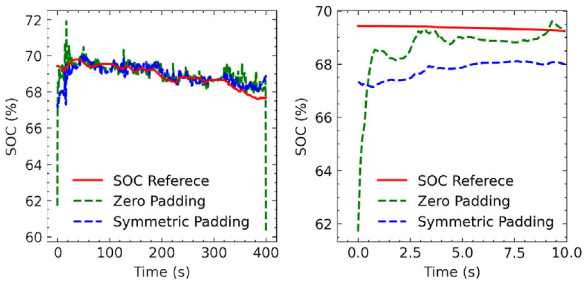


Fig. 5. SOC estimation results with two padding methods.

zero padding will only add zeros on both sides, regardless of the input data, while symmetric padding will mirror and inverse the input data. By this method we can guarantee the similarity of the padding data to the real data.

We greatly improved the estimation accuracy of SOC at the borders by using symmetric padding convolutional layer. Fig. 5 shows the SOC estimation results with two padding methods, respectively. We can see from the results that both methods have high and similar accuracy in the middle part of the estimation results. However, on the left and right sides, there is a significant error with zero padding, while using symmetric padding eliminates this error and makes the results have high accuracy at all points, including the starting point.

3.3. Total variation loss function

Neural networks have been used in a variety of battery scenarios, especially in SOC estimation, but one of the biggest drawbacks of neural networks is the uncertainty of the results. The data-based approach, due to its principle, leads to relatively independent and volatile output results from one another. As an evaluation metric for remaining battery capacity, SOC has temporal continuity, *i.e.,* the values of adjacent SOC values are very close to each other. Unlike model-based approaches, previous neural networks can't take advantage of this continuity, leading to the fluctuation in the results. There are many papers that address the problem of unstable output of neural networks by adding filters. Chen et al. [40] used a particle filter framework with grey neural network to predict remaining useful life of battery cells. Tian et al. [41] proposed a combined method for SOC estimation using LSTM and adaptive cubature Kalman filter. By adding a novel Kalman filter, their method exhibits excellent generalization ability and high accuracy. Chen et al. [28] proposed a synthetic SOC estimation method based on LSTM and adaptive H-infinity filter. In their conclusion, the maximum SOC error after filtering is only one-tenth of the original. These papers reduce the output volatility and improve the SOC estimation accuracy by introducing a post-processing process. These methods increase the complexity of the overall SOC estimation process and essentially perform output filtering to improve the smoothness of the estimation results. The filtering process is relatively independent of the neural network and has no effect on the accuracy improvement of the neural network itself.

Considering the limitation of the battery mechanism, the SOC variation rate of the battery should be kept within a limit. In order to take full advantage of temporal continuity of the SOC without increasing the complexity of neural network structure, we proposed a total variation loss function. Total variation was pioneered by Rudin, Osher, and Fatemi in 1992 to describe the amount of spurious details in the signal [42]. It is based on the principle that signals with excessive and possibly spurious detail have high total variation, that is, the integral of the absolute image gradient is high.

This algorithm was first used for image noise reduction, and compared to linear smoothing or median filtering, this method is very effective in preserving edges while smoothing out noise in flat areas even at low signal-to-noise ratios. For a one-dimensional discrete signal y_n, the total variation can be defined as:

$$V(y) = \frac{\sum_n |y_{n+1} - y_n|}{n} \tag{5}$$

From (5) we can see that the total variation is determined by the difference between adjacent values. We can add total variation as a penalty term to the loss function of our proposed method to help the optimizer update the parameters in the direction of smoother output. Generally neural networks take root mean square error (RMSE) or mean absolute error (MAE) as the loss function, we use the MAE here and the total loss function after adding the penalty term can be expressed by (6). *loss* is the total loss function of the training process. $loss_{mae}$ is the MAE loss and $loss_{tv}$ is the total variation loss. λ is the regularization weight, which determines the proportion of the total variation term. When $\lambda \to 0$, the smoothing effect is small and the output of the model fluctuates more. When $\lambda \to \infty$, the total variation term plays an increasingly important role, which will force the results to be smoother, or even output constant values. Therefore, the regularization weight has a significant impact on achieving proper smoothing.

$$loss = loss_{mae} + \lambda loss_{tv} \tag{6}$$

Previous SOC estimation methods, including SVM, LSTM, GRU, etc., cannot apply the total variation loss function because they can only estimate one value per inference. Our proposed SOC estimation method can easily suppress the random errors in the estimation results using total variation loss because all values are estimated at once. By adding penalty terms, we can significantly improve the model accuracy through optimization of the training process alone without changing the structure of the neural network. Fig. 6 shows the model validation results with 3 weighting factors.

From Fig. 6 we can get the following conclusions. When $\lambda = 0$, *i.e.,* no penalty term, the estimation result is around the true value, but there are obviously some fluctuations and the maximum error is large. When $\lambda = 50$, the fluctuation of the estimation result is suppressed and the result is around the true value, the maximum error is significantly reduced. When $\lambda = 500$, the estimation result is smoother but has deviated from the true value. Some estimation results even become all zeros due to excessive penalty. In the subsequent experiments we choose $\lambda = 50$ to balance the smoothing effect with the estimation accuracy. With this improvement we not only improve the stability of the model output, but also significantly reduce the maximum error in SOC estimation.

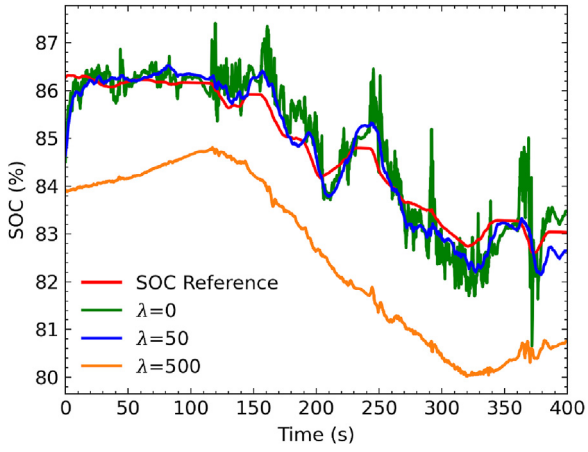


Fig. 6. SOC estimation results with 3 weighting factors.

3.4. Hyperparameter and training

Hyperparameters have a great impact on the network accuracy [43]. The number of convolutional layers, kernel size, layer number and other hyperparameters all affect the SOC estimation results. These hyperparameters can be selected over a wide range and together form a large search space. Neural network training is time-consuming, and using grid search would lead to high training cost, so we use Bayesian optimization algorithm to achieve hyperparameter optimization in a reasonable time [44]. Bayesian optimization can be summarized in a single sentence: a probabilistic model of the objective function is built and used to select the most promising hyperparameters to be evaluated in the true objective function. Optimal hyperparameters we selected are listed in Table 2. All models were trained on a 3090 GPU (24G memory) on a Windows 10 machine with TensorFlow 2.6.0 library.

Mean square error (MSE) is a common metric used in neural network training to measure model error. The change in MSE can indicate whether the model is converging and whether it is overfitting. Fig. 7 shows the changes in training MSE and validation MSE as the training process progresses. MSE decreases rapidly at the beginning of training and slows down afterwards, so we take the logarithm of the y-axis to make the results clearer. From the figure we can see that the training error and the validation error change very slowly after 80 epochs, so our training process stops at 100 epochs. The coefficient of determination of our model reached 0.997, which is a value very close to 1 and proves that our model has a high accuracy.

4. Results and discussion

4.1. Estimation under constant temperature

We used experiments data from all five available ambient temperature conditions except US06 drive cycles to train our SOC

Table 2
Hyperparameters of the best performing model.

Hyperparameters	Values
Kernel size	5
U-Net depth	3
Shrink ratio	5
Activation	ReLU
λ	50
Initial learning rate	1e-3

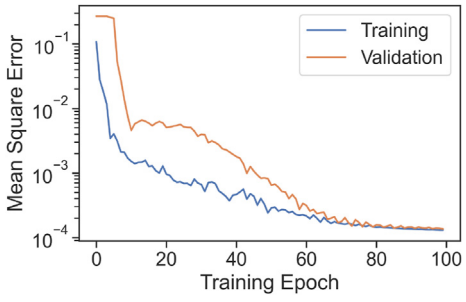


Fig. 7. Training MSE and validation MSE.

estimation network. The US06 drive cycles were selected as the validation data. The training time of the model is about 20 min. After the training, we used US06 drive profile to validate the SOC estimation results.

Our method can be used for both real-time SOC estimation and offline SOC estimation. In the case of real-time estimation, our method requires the same update calculation for each data point as the other NN based methods, with no significant difference in computational complexity. In the case of offline estimation our method has less computational complexity because all data can be estimated by a single calculation. The offline SOC estimation process is to feed the whole time series of battery data into our model and estimate the SOC of all data points at once, without the recurrent estimation process of RNN networks, which reduces the time-consuming. Since all the battery data were obtained offline in this validation, we used the offline method which is faster in computation. Figs. 8 and 9 show the SOC estimation results with errors for the US06 drive cycles at 25 °C and 0 °C ambient temperature, respectively. The estimation results have high accuracy under two different temperature conditions, where the MAE is 1.1%, RMSE is 1.4% and the max error is 7.4% at 25 °C, and the MAE is 0.9%, RMSE is 1.1% and the max error is 3.4% at 0 °C. These two results demonstrate the validity of out proposed method under constant temperature condition. With the total variation penalty term we improve the stability of the model, and the maximum error is greatly reduced compared to the model without penalty.

4.2. Estimation under variable temperature

Although the training data contains experimental data at five constant ambient temperature, the introduction of temperature data during the training process allows the model to gain generalization capability. We selected a set of variable temperature experimental data from the dataset, where the ambient temperature was gradually increased from 10 °C to 25 °C. The SOC estimation result of variable temperature is shown in Fig. 10. Throughout the discharge process, the temperature inside the chamber gradually increased from 10 °C to 23 °C, while the temperature of the battery gradually increased from 10 °C to a maximum of 25 °C. The MAE of SOC estimation is 1.0%, the RMSE is 1.3% and max error is 4.2%. Fig. 11 shows another set of data, in which the ambient temperature gradually increases from −20 °C to 10 °C in this experiment. During this process, the MAE is 1.2%, the RMSE is 1.5% and the max error is 5.0%. As the temperature increases, there are many temperature point data that are not available during training. The SOC estimation still has a high accuracy under the variable temperature condition, which proves the proposed model has the ability of generalization.

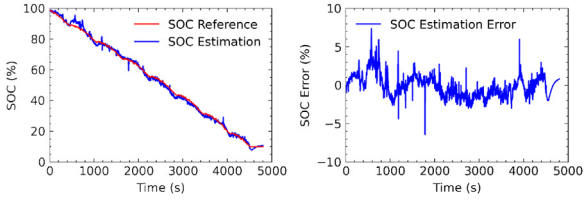


Fig. 8. SOC estimation result at constant ambient temperature (25 °C).

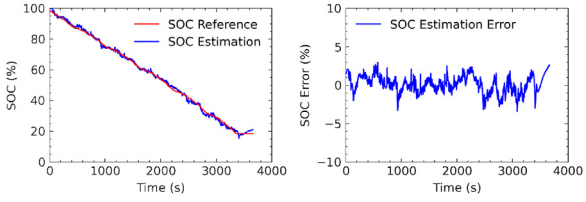


Fig. 9. SOC estimation result at constant ambient temperature (0 °C).

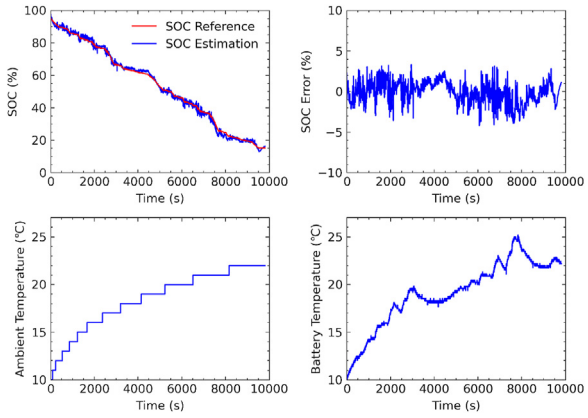


Fig. 10. SOC estimation result at variable ambient temperature (10 °C–25 °C).

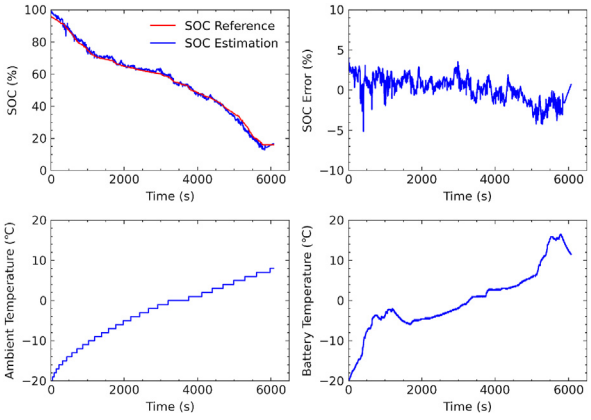


Fig. 11. SOC estimation result at variable ambient temperature (−20 °C to 10 °C).

4.3. Compare with EKF and RNN

In the previous section we have described the advantages of the proposed method without the need for a convergence process, and we performed a practical comparison here to illustrate this difference. In addition to the proposed method, we also used two other SOC estimation methods, extended Kalman filter (EKF) and RNN, both using the dataset introduced in the previous paper for parameter design and model training, and validated using US06 drive cycle. The focus of this paper is not on the design of EKF and RNN, so the estimation of these two methods does not reach state-of-the-art accuracy. This experiment is conducted mainly to compare the differences between the proposed method and the previous methods.

Fig. 12 shows the SOC estimation results of the three methods. The EKF method has a convergence process of length 400 s, which is not accurate for the first 400 s. Although the SOC initial error in Kalman filter method is not always large, this initial error is generally present and time is required to stabilize the estimation results. The RNN method needs sufficient length of data to start estimation due to the structure, so the first SOC estimation result is obtained at 400 s, before which the estimation can't be performed.

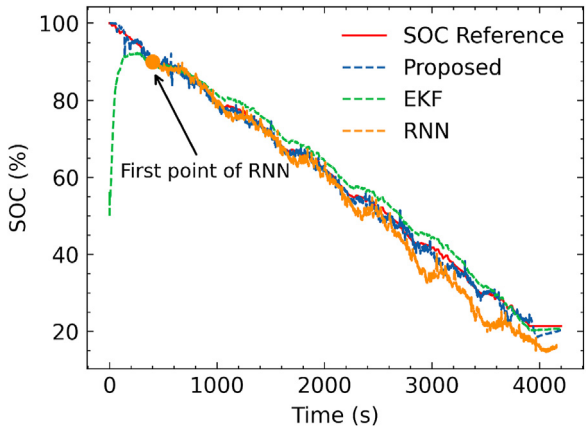


Fig. 12. SOC estimation results of three methods at constant ambient temperature (10 °C).

Using only convolutional layers makes the input and output lengths variable. Even if the input data is shorter than the training data, our proposed method still gives accurate estimation results, while other methods would fail to do so. 2) Our method performs SOC estimation for each data point of the input without convergence process, unlike RNN or EKF which require extra time to stabilize the output. 3) Through symmetric padding technique we improve the convolutional layers to solve the problem of large errors at the edges of the data in CNN. The SOC estimation has a consistent accuracy at the starting point with the later data. 4) Besides, we introduce the total variation penalty term in the training process, which greatly improves the smoothness of the output results and reduces the maximum error by optimizing the loss function alone without changing the model structure. The model trained using dynamic cycling data at five different temperatures also has high generalizability under variable temperature conditions. Our trained model has an RMSE of 1.4% for SOC estimation at constant temperature and 1.8% for SOC estimation at variable temperature.

Table 3
Estimation results of 8 variable ambient temperature conditions.

Profile	Ambient Start Temperature (°C)	Ambient End Temperature (°C)	MAE/RMSE (%) w/Temperature	MAE/RMSE (%) w/o Temperature
Cycle 1	10	23	1.2/1.5	1.4/1.8
Cycle 2	10	22	1.0/1.3	1.2/1.6
Cycle 3	10	22	1.5/1.8	1.1/1.4
Cycle 4	10	22	1.2/1.6	1.3/1.7
Cycle 1	−20	11	1.1/1.4	1.4/2.0
Cycle 2	−20	10	1.2/1.5	1.6/2.1
Cycle 3	−20	8	1.2/1.5	1.5/2.2
Cycle 4	−20	11	1.3/1.6	1.5/1.8

As a comparison, our proposed method can accurately estimate the SOC during the entire discharge process, including the data prior to 400 s, demonstrating our advantages.

4.4. Influence of temperature on SOC estimation

In our proposed model, we use current, voltage and temperature data as inputs, but in the actual electric vehicles there are few temperature sensors and most batteries do not have accurate temperature data. So we constructed a model with only current and voltage as inputs, and used trained the model with experiment data at five different ambient temperatures. The input data has only 2 dimensions, i.e., the normalized current and voltage. The SOC estimation results of the two models under variable temperature dynamic conditions are shown in Table 3. From the results we can see that our proposed model has high SOC estimation accuracy with or without introducing temperature as input. The estimation accuracy is reduced when ignoring the temperature data, but still within acceptable range using this dataset. This also demonstrates the adaptability of the proposed method to different data, and this structure can automatically extract valid information from multiple types of data. The accuracy of the proposed method for SOC estimation can be further improved with the increase of data collected by the BMS, such as mechanical stress, and heat flow density information. Using multiple auxiliary information to improve the accuracy of battery state estimation is our next research goal.

5. Conclusion

In this paper, we present a CNN-based SOC estimation neural network. This model has a U-Net architecture, consisting of convolutional techniques, without recurrent neural network structures or fully connected layers. The main contributions as follows: 1)

Credit author statement

Xinyuan Fan: Conceptualization, Software, Writing - Original Draft. Weige Zhang: Supervision, Writing - Review. Caiping Zhang: Funding acquisition, Project administration. Anci Chen: Data Curation, Validation. Fulai An: Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported in part by the National Natural Science Foundation of China (No. 51977007), in part by Beijing Natural Science Foundation (NO. 3212033).

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